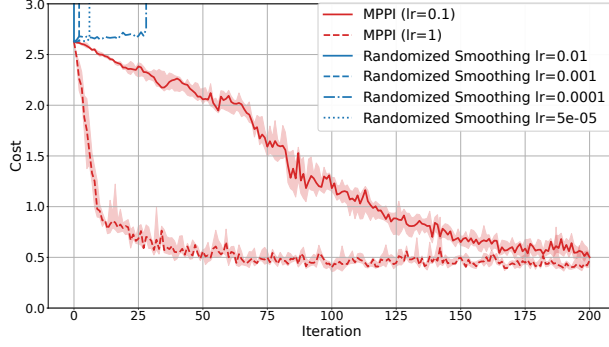
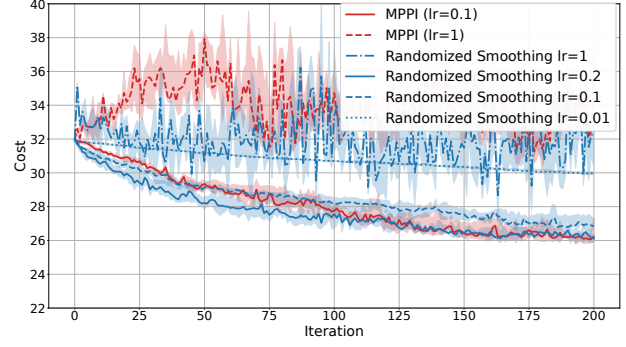


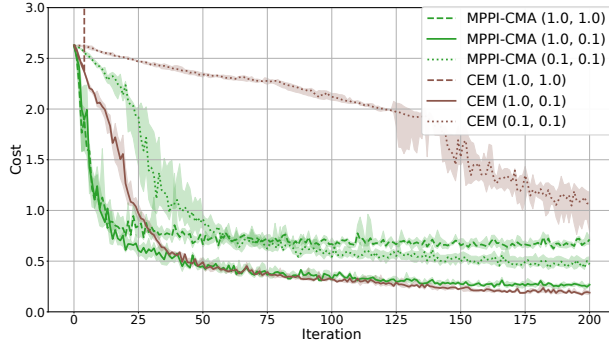


(a) Go2Walk

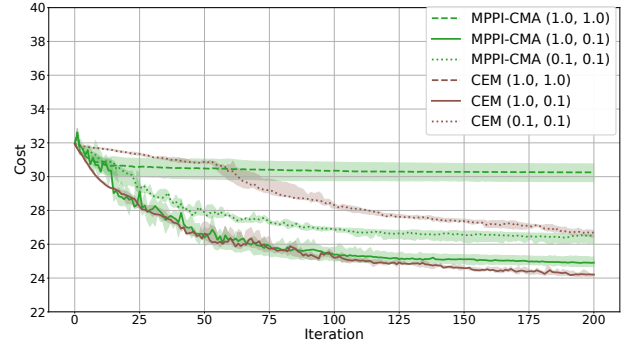


(b) HumanoidStandup

Figure 1: Vanilla methods need learning-rate tuning: the best MPPI learning rate is task-dependent (vanilla $lr=1$ is best on Go2Walk but unstable on HumanoidStandup, where $lr=0.1$ is needed), and Randomized Smoothing’s best LR likewise varies by task (it diverges at every LR on Go2Walk).

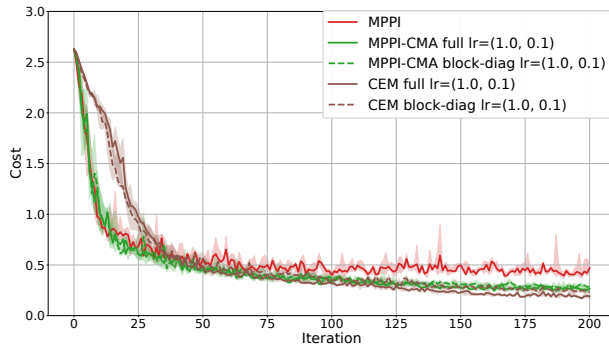


(a) Go2Walk

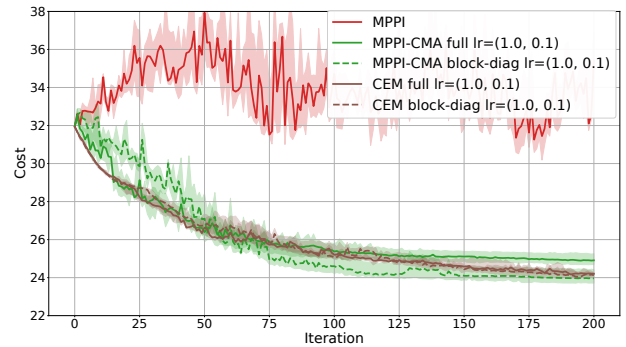


(b) HumanoidStandup

Figure 2: Covariance adaptation needs two distinct learning rates: for both MPPI-CMA and CEM, $(\text{mean}, \text{cov}) = (1, 0.1)$ outperforms a single shared rate $(1, 1)$ or $(0.1, 0.1)$ on both Go2Walk and HumanoidStandup.



(a) Go2Walk



(b) HumanoidStandup

Figure 3: Covariance adaptation helps and block-diagonal $\approx$ full: MPPI-CMA and CEM (full and block-diagonal) improve over the MPPI baseline on both Go2Walk and HumanoidStandup; block-diagonal closely matches full covariance.

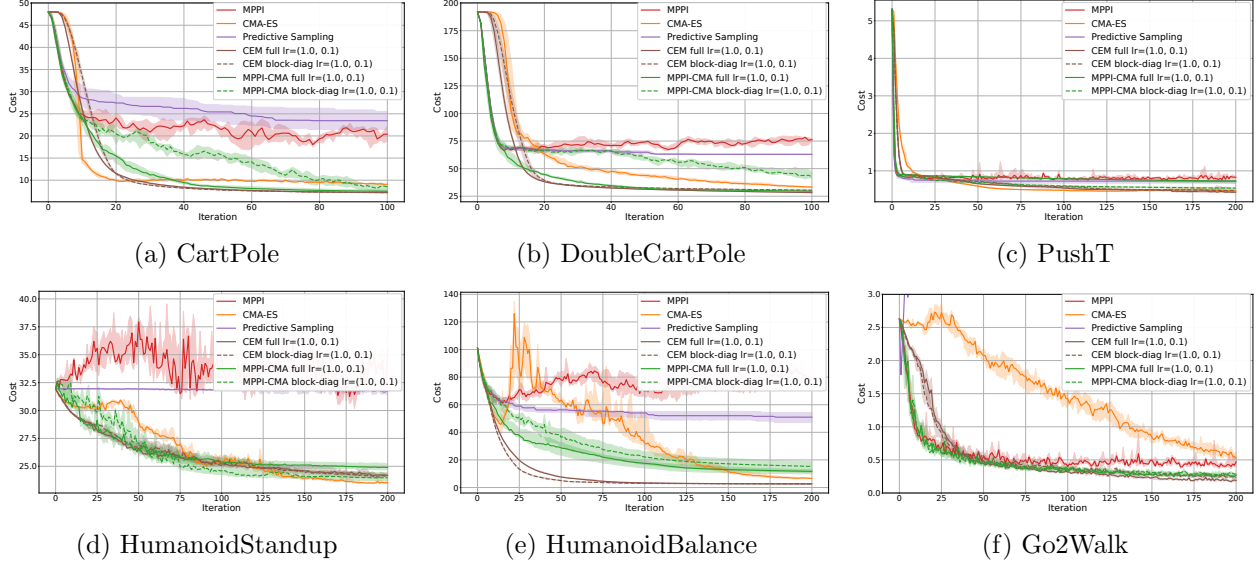


Figure 4: Per-task convergence (appendix, full 7-algorithm set): MPPI, CMA-ES, Predictive Sampling, CEM (full and block-diagonal), and MPPI-CMA (full and block-diagonal), all at $lr = (1.0, 0.1)$ where applicable; colour denotes the algorithm, dashed = block-diagonal covariance.